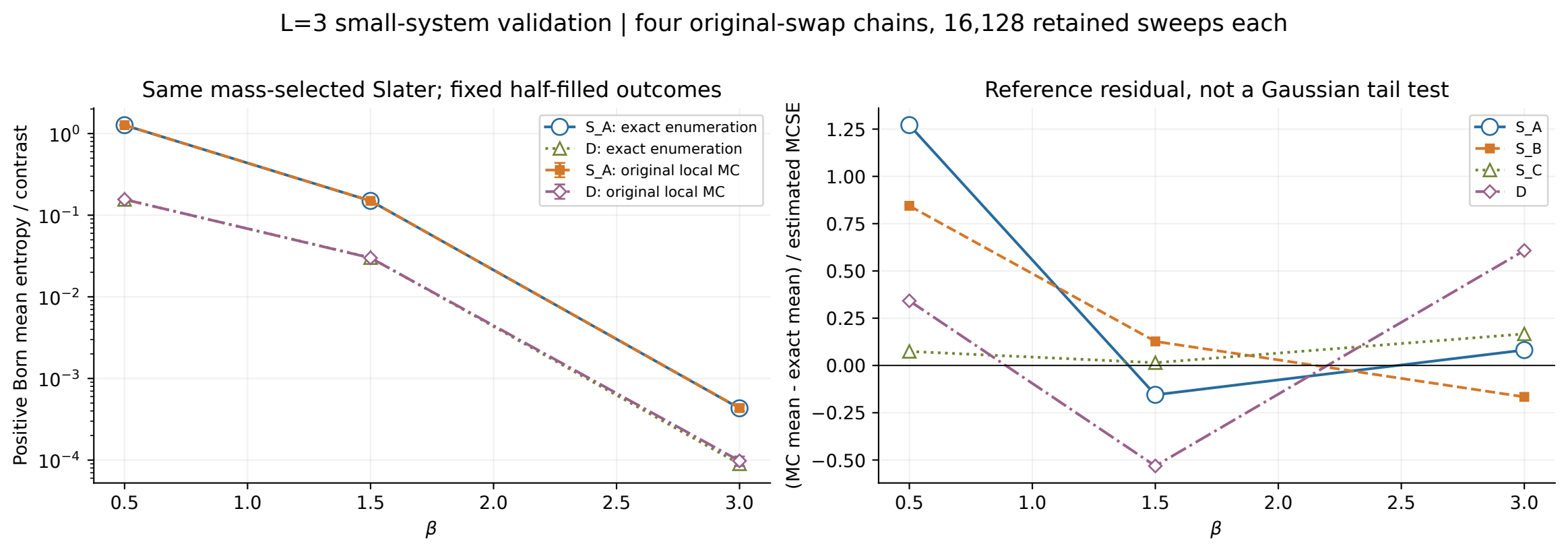
L=3 small-system validation | four original-swap chains, 16,128 retained sweeps each
Same mass-selected Slater; fixed half-filled outcomes
Positive Born mean entropy / contrast
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0.5
1.0
1.5
2.0
2.5
3.0
β
S_A: exact enumeration
D: exact enumeration
S_A: original local MC
D: original local MC
Reference residual, not a Gaussian tail test
(MC mean - exact mean) / estimated MCSE
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
0.5
1.0
1.5
2.0
2.5
3.0
β
S_A
S_B
S_C
D
Exact enumeration has no MC error. Error bars retain each sample's A/B/C covariance; D=(2SA+SC)/3-SB.
Small absolute error at beta=3 is compatible with sizable relative error; this check does not establish large-L convergence.